

Increments and Differentials.

(83)

Consider the function $y=f(x)$.

If x changes from x_1 to x_2 , then the amount of change is often denoted by the symbol Δx , that is

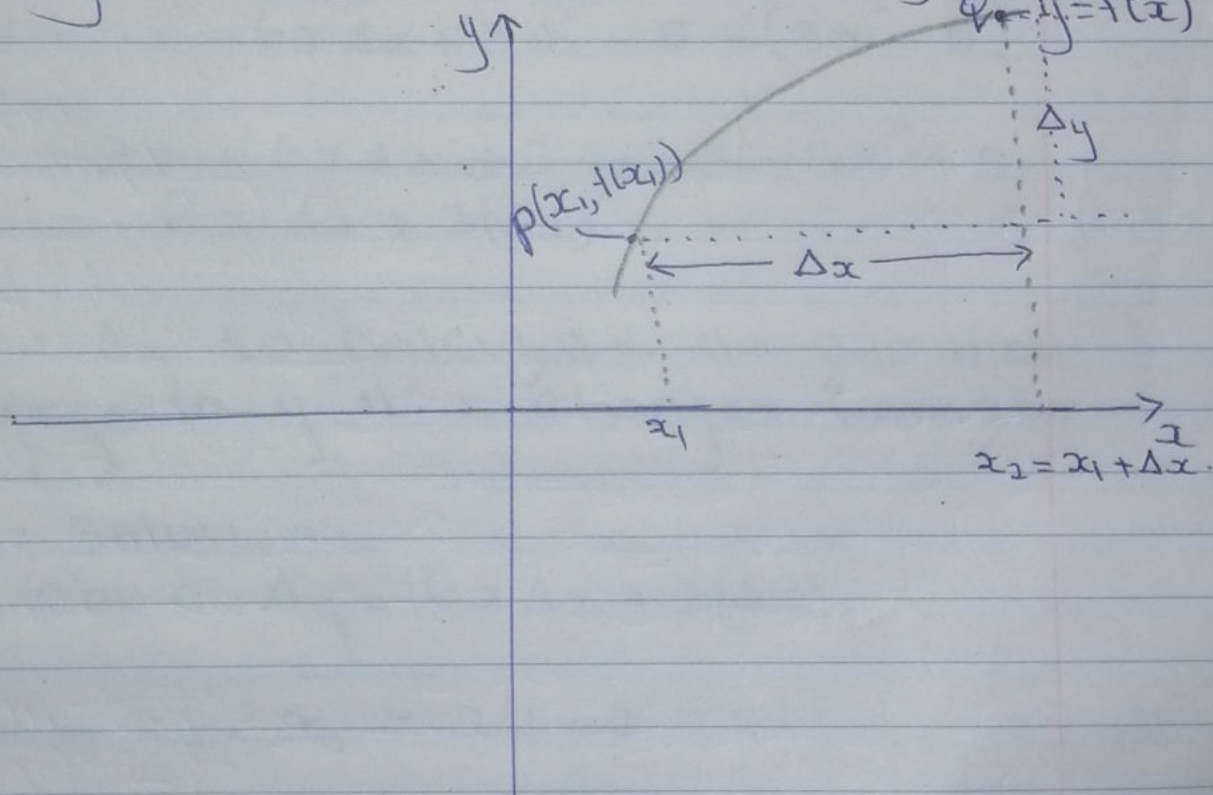
$$\Delta x = x_2 - x_1$$

The number Δx is called an increment of x . Notice that

$$x_2 = x_1 + \Delta x.$$

Similarly, Δy will denote the change in the dependent variable y that corresponds to the change Δ . Thus

$$\Delta y = f(x_2) - f(x_1) = f(x_1 + \Delta x) - f(x_1).$$



Where the coordinate of q is $(x_1 + \Delta x, f(x_1 + \Delta x))$

Definition

Let $y = f(x)$, where f is a function. If x is given an increment Δx , then the increment Δy of y is

$$\Delta y = f(x + \Delta x) - f(x)$$

Example: Suppose $y = 3x^2 - 5$.

(a) If x is given an increment Δx , find Δy .

Solution

Let $f(x) = 3x^2 - 5$.

$$\begin{aligned}\Rightarrow f(x + \Delta x) &= 3(x + \Delta x)^2 - 5 \\ &= 3x^2 + 6x\Delta x + 3\Delta x^2 - 5.\end{aligned}$$

$$\begin{aligned}\Delta y &= f(x + \Delta x) - f(x) \\ &= 3x^2 + 6x\Delta x + 3\Delta x^2 - 5 - (3x^2 - 5) \\ &= 3x^2 + 6x\Delta x + 3\Delta x^2 - 5 - 3x^2 + 5 \\ &= 6x\Delta x + 3(\Delta x)^2\end{aligned}$$

(b) Use Δy to calculate the numerical change in y if x changes from 2 to 2.1.

Solution

from above $\Delta y = 6x\Delta x + 3(\Delta x)^2$.

$$\Delta x = x_2 - x_1 = 2.1 - 2 = 0.1.$$

$$\begin{aligned}\Rightarrow \Delta y &= 6x\Delta x + 3(\Delta x)^2 \\ &= 6 \times 2 \times 0.1 + 3(0.1)^2 \\ &= 1.2 + 0.03 = 1.23.\end{aligned}$$

(85)

Thus, y changes by an amount 1.23 if x changes from 2 to 2.1.

We could also find Δy directly as follows;

$$\begin{aligned}\Delta y &= f(x + \Delta x) - f(x) = f(2.1) - f(2) \\ &= 3(2.1)^2 - 5 - [3(2)^2 - 5] \\ &= 1.23\end{aligned}$$

Recall that

$$f'(x) = \lim_{\Delta x \rightarrow 0} \frac{f(x + \Delta x) - f(x)}{\Delta x} = \lim_{\Delta x \rightarrow 0} \frac{\Delta y}{\Delta x}.$$

$\Rightarrow$ If $f'(x)$ exists, then

$$\frac{\Delta y}{\Delta x} \approx f'(x) \text{ if } \Delta x \approx 0$$

We can also write

$$\Delta y \approx f'(x) \Delta x \text{ if } \Delta x \approx 0$$

Definition

Let $y = f(x)$, where f is differentiable, and let Δx be an increment of x .

- i. The differential dx of the independent variable x is $dx = \Delta x$.
- ii. The differential dy of the dependent variable y is

$$dy = f'(x) \Delta x = f'(x) dx.$$

$$\Rightarrow \Delta y \approx dy = f'(x) dx = (D_x y) dx.$$

Example 2.

(86)

If $y = 3x^2 - 5$, use dy to approximate Δy if x changes from 2 to 2.1.

Solution

If $f(x) = 3x^2 - 5$, then from our previous example, we have that when x changes from 2 to 2.1, then

$$\Delta y = 1.23.$$

Using the differentials we have

$$dy = f'(x)dx = 6x dx.$$

In this present example,

$$x = 2, \Delta x = dx = 0.1, \text{ and thus}$$

$$dy = 6 \times 2 \times 0.1 = 1.2.$$

$\Rightarrow$ Our approximation is correct to one decimal place.

Example 3:

If $y = x^3$ and Δx is an increment of x , find (a) Δy .

Solution

$$\Delta y = f(x + \Delta x) - f(x) = (x + \Delta x)^3 - x^3.$$

$$= x^3 + 3x^2\Delta x + 3x\Delta x^2 + \Delta x^3 - x^3$$

$$= 3x^2\Delta x + 3x\Delta x^2 + \Delta x^3$$

(b) dy :

$$\text{By definition } dy = f'(x)dx = f'(x)\Delta x = 3x^2\Delta x.$$

(c) $\Delta y - dy$.

$$\Delta y = 3x^2\Delta x + 3x\Delta x^2 + \Delta x^3$$

$$dy = 3x^2\Delta x$$

$$\Rightarrow \Delta y - dy = 3x\Delta x^2 + \Delta x^3 = 3x(\Delta x)^2 + (\Delta x)^3.$$

(d) the value of $\Delta y \approx dy$ if $x=1$ and $\Delta x = 0.02$. (87)

Solution

$$\begin{aligned}\Delta y - dy &= 3x^2 \Delta x + \Delta x^3 \\ &= 3 \times 1 \times (0.02)^2 + (0.02)^3 \\ &= 3 \times 0.0004 + 0.000008 = 0.001208\end{aligned}$$

Example 4:

Use differentials to approximate the change in $\sin \theta$ if θ changes from 60° to 61° .

Solution

If $y = \sin \theta = f(\theta)$, then

$$dy = f'(\theta) d\theta = \cos \theta d\theta$$

$$\Rightarrow dy = f'(\theta) d\theta = \cos \theta \Delta \theta$$

$$= \cos \theta \cdot 1^\circ$$

But $60^\circ = \pi/3$ and $1^\circ = \frac{\pi}{180}$

$$= dy = \cos(\pi/3) \cdot \left(\frac{\pi}{180}\right) = \frac{1}{2} \times \frac{\pi}{180} \approx 0.0087$$

$$\approx 0.0087.$$

Using a Calculator, we find that to four decimal places

$$\begin{aligned}\Delta y &= \sin 61^\circ - \sin 60^\circ \approx 0.8746 - 0.8660 \\ &= 0.0086.\end{aligned}$$

Hence the error involved in using dy to approximate Δy is roughly 0.0001.

Example 5

(88)

How fast is area A of a circle increasing with respect to its radius when the radius is 5m?

Solution

The rate of change of the area with respect to the radius is

$$\frac{dA}{dr} = \frac{d}{dr} (\pi r^2) = 2\pi r.$$

Where $r = 5\text{m}$, the area is changing at the rate of $2\pi(5) = 10\pi \text{ m}^2/\text{min}$.

Example 6

Suppose that the temperature at a certain location t hours after noon on a certain day is $T^\circ\text{C}$, where

$$T = \frac{1}{3}t^3 - 3t^2 + 8t + 10 \quad \text{for } (0 \leq t \leq 5).$$

How fast is the temperature rising or falling at 1:00pm? at 3:00pm? At what instants is the temperature stationary?

Solution

The rate of change of the temperature is $\frac{dT}{dt} = t^2 - 6t + 8 = (t-2)(t-4)$

* If $T=1$, then $\frac{dT}{dt} = 3$, so the temperature is rising at the rate of 3 degrees C/h at 1:00pm

* If $T=3$, then $\frac{dT}{dt} = -1$, so the temperature is falling at the rate of 1 degree C/h at 3:00pm

(89)

- * The temperature is stationary when
 $\frac{dT}{dt} = 0 \Rightarrow (t-2)(t-4) = 0$
 $\Rightarrow t=2$ and $t=4$
 $\Rightarrow$ at 2:00 pm and 4:00 pm.

Example 7

A spherical tank to hold hydrogen is designed with a radius of 6m. The allowable error in the actual radius is ± 2 cm. Find the approximate excess capacity the tank could have over its design volume.

Compare this with the exact value of the maximum excess capacity.

Solution

The volume $f(r)$ of a sphere is given by $V = \frac{4}{3}\pi r^3$. We take $r=6$ m

and $\Delta r = dr = 0.02$. Since $2\text{ cm} = 0.02\text{ m}$.

The approximate change in volume if the radius is 6.02 instead of 6 is

$$\begin{aligned} dv &= f'(r)dr = f'(r)\Delta r = 4\pi r^2 \Delta r \\ &= 4\pi(6^2) \times 0.02 \\ &= 9.047786 \\ &\approx 9.05\text{ m}^3. \end{aligned}$$

The exact change is

$$\begin{aligned} \Delta v &= f(r+\Delta r) - f(r) \\ &= f(6.02) - f(6) \\ &= \frac{4}{3}\pi(6.02)^3 - \frac{4}{3}\pi 6^3 \approx 9.078\text{ m}^3. \end{aligned}$$

(90)

If we know the value of a function f at a given point x , then we can get its value at a nearby point $x + \Delta x$ by adding the change Δy to the original value; that is

$$f(x + \Delta x) = f(x) + \Delta y.$$

Furthermore, if $f'(x)$ exists, we can approximate Δy by dy if we take $dx = \Delta x$

$$\Rightarrow f(x + \Delta x) = f(x) + dy.$$

or

$$\begin{aligned} f(x + \Delta x) &= f(x) + f'(x) dx \\ &= f(x) + f'(x) \Delta x. \end{aligned}$$

This result is known as linear approximation.

Example: 8

Use linear approximation to approximate $\sqrt[3]{65}$

Solution

$$\text{Let } f(x) = \sqrt[3]{x} = x^{1/3}.$$

The point nearest to 65 at which we know the value of f is $x = 64$,
Since $\sqrt[3]{64} = 4$.

$\Rightarrow$ In our equation we take $x = 64$ and $\Delta x = 1$.

$$f(x) = x^{1/3} \Rightarrow f'(x) = \frac{1}{3} x^{-2/3}.$$

$$\begin{aligned} \Rightarrow f(65) &= f(64) + f'(64) \times 1 \\ &= 4 + \frac{1}{3} (64)^{-2/3} \approx 4.0208. \end{aligned}$$

Exercise:

(91)

1. A projectile fired at an angle θ from the horizontal with an initial velocity of V_0 meters per second will rise to a maximum height of

$$y = \frac{V_0^2 \sin^2 \theta}{2g}$$

where $g = 9.80 \text{ m/sec}^2$ is the acceleration due to gravity. Approximately how much higher will a projectile fired at an angle of 31° go than one that is fired at an angle of 30° , if the initial velocity in each case is 80 m/sec ?

2. The side of a square is measured to be 18 cm . If there is a possible error in the measurement of 0.05 cm , find the approximate possible error in
- (a) the area.

(b) the perimeter.

3. Use differentials to approximate $(0.98)^4$.

Hint: (let $y = f(x) = x^4$ and consider

$$f(x + \Delta x) = f(x) + \Delta y,$$

where $x = 1$ and $\Delta x = -0.02$.)

What is the exact value of $(0.98)^4$?

(92)

Revision Questions on Derivatives

4. Find the tangent and normal lines to the curve $y = \tan \frac{\pi x}{4}$ at the point $(1, 1)$.

5. Find $\frac{dy}{dx}$ in each case.

(a) $x^3 - xy + y^3 = 1$

(b) $x^2y + xy^2 = 6$

(c) $y^3 + y = 2 \cos x$

(d) $y = \ln(x^2 + 3x + 1)$

(6) Find the tangent line to the curve

(a) $x^3 + y^2 = 2$ at the point $(1, 1)$

(b) $x^{\frac{3}{2}} + 2y^{\frac{3}{2}} = 17$ at $(1, 1)$

(c) $x^3y^3 + y^2 = x + y$ at $(4, 1)$

(d) $x + \sqrt{xy} = 6$ at $(4, 1)$

(7) Given that $xy + x - 2y - 1 = 0$, find $\frac{dy}{dx}$ and $\frac{d^2y}{dx^2}$.

(8) If $y = \ln(x + \sqrt{1+x^2})$, show that

$$y' = \frac{1}{\sqrt{1+x^2}}$$